

Here is the strategic summary of our bidding plan for Round 2, formatted for you to share with your teammates.

STRATEGIC MEMORANDUM: ROUND 2

AUCTION THEORY

Subject: Optimizing the 25% Market Volume Bid for Survival & Net PnL

1. THE OBJECTIVE

To secure the 25% trading volume boost while protecting our ability to reach the **200,000 Xirec Net PnL** threshold. Because the leaderboard resets after 200k, every Xirec earned above that mark has zero utility. Our bid is an **insurance premium**, not an investment for infinite growth.

2. AUCTION MECHANICS & CONSTRAINTS

- **First-Price Auction:** We pay exactly what we bid. There is no "market price" for the volume; if we bid 50,000, we pay 50,000.
- **Net PnL Settlement:** The bid is deducted from our final total.
 - *Critical Risk:* A high bid moving our "effective" goalpost. If we bid 20k, our algorithm must now generate 220k just to hit the 200k survival line.
- **Winner Pool:** The top 50% of bidders win. Out of ~22,300 teams, we only need to outbid the **11,151st team**.

3. GAME THEORY ANALYSIS

The "Median" Participant

We are currently ranked in the Top 6% (1,189th). We are significantly more efficient than the median team.

- **The Inactive Floor:** Thousands of registered teams are "ghosts" or have zero PnL. They will bid 0 or 1.
- **The Rational Ceiling:** A team ranked 10,000th might only expect to make 5,000 Xirec total. They cannot mathematically bid more than 5,000 without guaranteed failure. By bidding slightly above the *median's* maximum capacity, we guarantee a win without paying "Elite" prices.

The "Winner's Curse" in Net PnL

In a "Pay-What-You-Bid" system, the "Winner's Curse" occurs when you win the auction but pay

so much that the cost exceeds the benefit.

- If the 25% volume boost grants us an extra 10,000 Xirec, but we bid 15,000 to "ensure" we get it, we have actively harmed our chances of hitting 200,000.

The Discord Leak Factor

While many teams now have access to "leaked" strategies (e.g., Pepper Root holding), this simply raises the **baseline PnL**—it does not necessarily raise the **bidding aggression**. Many teams will be conservative with their bids to ensure they don't fall below the 200k net mark.

4. BIDDING STRATEGY

A. The "Insurance Premium" Approach

We should not bid based on our *total* profit, but rather on the **marginal gain** provided by the 25% extra volume.

- **Formula:** Bid = (Expected Gain from Boost) * Risk Factor
- Given the 50% success rate, a high-conviction bid of 80% of our gains is unnecessary and dangerous. A bid of **5-15%** of our marginal gain is likely sufficient to clear the median.

B. Exploiting Psychological Anchoring

Participants gravitate toward round numbers (100, 500, 1,000, 5,000).

- **Tactical Execution:** Use "Weird Numbers." Bidding **1,537** instead of **1,500** allows us to leapfrog every team anchored to that round number for a negligible cost.

5. SUMMARY RECOMMENDATION

We recommend a **Low-to-Mid Tier Safety Bid** (estimated between **2,000 – 5,000 Xirec**).

- This bid is high enough to comfortably outpace inactive teams and low-efficiency bots.
- It is low enough that it does not significantly jeopardize our 200k Net PnL target if market conditions shift.
- **Goal:** Be the 5,000th highest bidder, not the 1st. There is no prize for overpaying.